IS450 Project Proposal

Title: Text Mining Analysis in the Financial Technology Domain

Section G2

Group 6

Members

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1. Introduction to the Business Domain and Problem

Investment analysis requires synthesizing information from various sources such as financial statements, earnings calls, analyst reports and news. Investors also need to account for inter-company dependencies such as supplier, customer or partners. In practice, this information is highly fragmented, large in volume and time-sensitive which makes holistic assessment difficult. As such, investors may miss early signals about a firm's financial health and overlook risks or opportunities emerging from related firms in its supply chain. Our project aims to solve this problem by building a text-mining pipeline that provides a 'bird's eye' view of a company's financial health and potential growth to reduce cognitive overload while preserving informational accuracy.

2. Data Statistics and Text Challenges

2.1 Data Overview

We will be selecting the earning reports (EDGAR SEC 10Q / 10K filings) of publicly listed technology companies between 2008 to 2023. This allows us to reduce cross-industry noise, while ensuring that data reflects current market conditions thereby improving comparability.

2.2 Data Statistics

Our dataset would consist of various financial texts such as quarterly press releases, news from reliable sources that meet a baseline of reliability and earning call transcripts, which helps to increase the quality of analysis. This data is released on a quarterly basis or event-driven basis so there is uneven document frequency across companies and time periods.

2.3 Text Challenges

1. **Heterogeneous Data Format**: Financial texts often have a mix of numerical numbers and text which may be difficult to jointly model quantitative and qualitative information
2. **Domain-specific Language**: Specialised terminology are not understood by general models
3. **Speculative Language:** Earning calls usually contain projections which will interfere with sentiment analysis
4. **Divergent Stakeholder Narratives**: Official financial statements may be overly optimistic, and can end up contradicting the interpretations of the broad market. This can potentially lead to inaccurate sentiment analysis

3. Text Mining Tasks

3.1 Task 1: Sentiment Analysis

3.1.1 Business Benefits

Our sentiment analysis intelligence platform helps firms minimize **Time-to-Insight, while maximizing Research Throughput Rate**. Sentiment signals can be used as part of portfolio management techniques to gain first mover advantage, potentially helping analysts to achieve investments returns exceeding market benchmarks consistently.

3.1.2 Approach

To demonstrate the value of sentiment analysis in algorithmic trading, portfolio optimization and stock price forecasting, we are planning to develop a 2 stage workflow. First we focus on accurately classifying aspect-based sentiment *(positive, neutral, negative - multiclass classification)*. Second, we integrate these signals into *price forecasting models and / or portfolio optimizers to quantify the performance improvement compared to a baseline model without sentiment features*.

1. **Text Preprocessing:** Extract each subsection of 10Q reports (e.g. MD&A , Risk Factors etc) + POS Tagging, Stop word removal, Lemmetization etc
2. **Aspect Extraction:** Utilizing Gliner (Generalist and Lightweight NER via Zero Shot) for each subsection, e.g. *'Revenue Growth', 'Supply Chain', 'Cost'* etc
3. **Ground Truth Labeling:**
    a. We will first manually label 1000+ aspect-level sentences from 10-Q reports as positive/neutral/negative according to domain knowledge annotation guidelines.
    b. Subsequently we will make use of a LLM to conduct few-shot-prompting to label the rest of our data
    - A random subset (10-15%) of LLM-generated labels will then be manually reviewed to ensure accuracy and alignment with our annotation guidelines
4. **Model Experimentation + Evaluation I**
    a. **Out-of-Time Split:** We will conduct a time-based split to partition our dataset into training, and holdout test sets to train and evaluate our models, preventing data leakage
    b. **Rule-Based Approach:** Loughran-McDonald Dictionary
    c. **Traditional ML:** SVM (Support Vector Machine) / Naive Bayes Classifier
    d. **Transformer Approach:** FinBert

e. **Evaluation:** We will then identify model overfitting and underfitting issues on holdout validation set / Conduct K-Fold Cross Validation. The metrics that we would be using include *Precision, Recall, F1 Score, Macro F1, Weighted F1*

5. **Hyperparameter Tuning:** Enhance model accuracy via approaches such as RandomSearchCV / Optuna
6. **Model Evaluation II + Final Model Selection:** Evaluation on Hold-Out Test Set
7. **Class-to-Score Conversion:** $P(Positive) - P(Negative)$ *(Sentiment Scores to act as input into end data products)*

3.1.3 End Data Product Ideas (Sentiment-Analysis Module)

- **Company Sentiment Intelligence Dashboard**
  - **Aspect-Based Sentiment Breakdown:** Multi-dimension radar chart (Financial, Legal, Operations, Liquidity, Related Companies etc) showing respective sentiment scores
  - **Risk Signal Extraction:** Top-*K* most positive & negative sentences, with their scores in 10Q reports / analyst reports
  - **Related Entity Sentiment:** Displays aggregated sentiment scores of related companies (e.g. suppliers, partnerships, competitors)

- **Sentiment-Based Portfolio Optimization Tool:**
  - **Interactive Portfolio Optimizer:** User-adjustable inputs (sentiment-threshold, position limits) with Sharpe Ratio maximization outputting optimal portfolio allocation, expected returns and max drawdown
  - **Portfolio Optimizer Algorithm Backtest Check:** Historical Simulation Check (for analysts to check if sentiment is aligned with price data)

- **Sentiment Trends & Early Warning System**
  - **Historical Sentiment Trend Chart:** Shows how company sentiment changed over past few years
  - **LSTM, ARIMA, Prophet Model Price Forecaster Buttons**
    - **Input:** Sentiment Time Series Features & Historical Prices
    - **Output:** Future price forecast, with confidence score
  - **Price Forecasting Algorithms' Backtest Check:** Historical Simulation Check (for analysts to check if sentiment is aligned with price data + whether they should use this model as trading strategy)

3.2 Task 2: Named Entity Recognition + Relation Classification

3.2.1 Business Benefits

Identifying the companies within a firm's value chain such as suppliers, customers or investors gives investors a clearer picture of the business ecosystem. Investor involvement can signal growth potential while reliance on key suppliers can arise operational risks. This information can help investors assess both opportunities and vulnerabilities more effectively.

3.2.2 Approach

1. **Name Entity Recognition**: To support downstream relationships identification
   a. FinBERT as a base model with token-classification (NER) head to extract company name. This step applies BIO tagging, then merge ORG tokens to get the complete company name from articles that we feed into the model.
      i. Evaluation: Calculate precision/recall/F1-score against a manually annotated dataset
2. **Entity Normalisation**: resolve extracted company mentions into canonical company names to ensure consistency across articles (Google LLC", "Alphabet Inc." = "Google")
   a. Text processing: lowercase, punctuation removal, legal suffixes removal
   b. Python fuzzymatch to match words: If mentions are under confidence threshold, then we will manually resolve
      i. Evaluation: Coverage, the proportion of mentions that are successfully normalised
3. **Candidates Generation**: find companies that that co-occur in the same sentence (i.e. {Company A, Company B, sentence}. Sentences that only have 1 company mentioned will be dropped.
   a. Evaluation: Coverage: the proportion of company names that have been correctly normalised
4. **Generate labelled training data** (i.e. {Tesla, BYD, 'competitor'}) and labels (i.e (investor, supplier, customer, competitor) using LLM then perform manual spot-checking to correct obvious errors
5. **Relation Classification**
   a. Split: time split - train on older article, test on newer article
   b. Baseline Model: Naives Bayes
      i. TF-IDF: Sentence, relation marker
      ii. Entity order: binary feature to indicate if Company X is in front of Company Y (i.e. X invests in Y)
      iii. Relationship marker: generate list of key words per relationship types ({investor: (invested, funding, stake)}
   c. Deep Learning model: FinBERT
      i. Features: Mark the 2 target companies using special entity tokens to focus model attention.
6. **Evaluation**:
   a. Confusion Matrix Analysis (Accuracy, Precision, Recall, F1-score)

b. Rank the extracted relations by model confidence then measure precision among the top K predictions

3.3.3 End Data Product Ideas (NER + RE Module)

- Core Idea:
  - Input: A finance article
  - Output: Company X, Company Y, their relationship
- End product: Financial Knowledge Graph
  - This graph will be on an interactive web interface that allows users to query a company and explore its financial ecosystem. Nodes represent companies, edges represent type of relationship and edge direction represents fund flows. (Refer to Figure 3 in Appendix)

3.3 Task 3: Topic Modelling

3.3.1 Business Benefits

Topic modelling would allow for analysts to **identify potential risks or opportunities based on surfacing topics** during critical events (e.g., 8-K filings), which often signal emerging financial concerns or strategic shifts in a company's direction that can impact the company's investment outlook. By tracking topics over time, analysts would be able to identify narrative changes, place more resources into relevant documents, and validate their findings through supporting evidence. Topic modelling can also help to enable **improved categorisation and indexing** by improving search consistency and retrieval speed of evidence and documents, facilitating for **faster, automated workflows**.

3.3.2 Approach

Our team has devised the following NLP Pipeline in order to achieve the Topic Modelling functionality within our project:

1. **Pre-processing:** Tokenisation, Stopword Removal, Lemmatisation, Phrase Detection will be incorporated to preserve financial semantics, chunking of large documents etc.
2. **Representation of Text:** Bag of Words Representation (TF-IDF), Embedding (BERT/finBERT)
3. **Discover Topics:** Bag of Words Representation (NMF, LDA), Clustering (BERTopic/Top2Vec/Mamba SSM)
   → Our team plans on experimenting with the different models available (non-exhaustive). Mamba SSM is of particular interest due to the purported ability to remember longer contextual information better when faced with long documents, potentially allowing for more accurate topic modelling.
4. **Output/Presentation:** Table with topics along with supporting documents, trend lines
5. **Evaluation:**
   a. **Coherence Score:** Measures whether words within a topic are semantically related and make sense together
   b. **Topic Diversity:** Measures how distinct each topic is from the others, ensuring minimal overlap
   c. **Topic Consistency:** Documents are matched to the same topics through different runs
   d. **LLM-As-A-Judge:** When faced with the same document, we will prompt various LLMs to provide the categorisation, along with an evaluation of our own model.
   e. **Manual Evaluation:** Manual intervention by humans to identify if a particular document is correctly identified and coherent

3.3.3 End Data Product Ideas (Topic Modelling Module)

- **Company Summary**
  - Provides overview of the company by combining results from Sentiment Analysis and NER
  - Includes a consolidated overview of a company's key discussion themes within a specified time frame
- **Topic Trend Line**
  - Given a company name, show the top 5 commonly discussed topics and the prevalence
  - Upon clicking on topic, provide more information and the supporting documents
  - Provide commonly discussed topics within the industry, and a comparison between the industry and selected company
- **Search and Indexing**
  - Enable analysts to retrieve documents associated with specific topics and relevance scores.
  - Filters by company, time, topic etc.
- **Emerging Topic Alert**
  - Allows analyst to "watch" topics or companies
  - Upon increase in mentions about topic or other thresholds set, alert analyst

Appendix

Desired Outputs for Sentiment Intelligence Platform (Fig 1 &2)

```
text

┌──────────────────────────────────────────────────────────┐
│ TESLA INC. (TSLA) - Q3 2024 Sentiment Report             │
├──────────────────────────────────────────────────────────┤
│                                                          │
│ Overall Sentiment Score: -0.25 🟠 (Slightly Negative)    │
│                                                          │
│ 📊 ASPECT-BASED BREAKDOWN:                               │
│ ┌──────────────────┬──────────┬──────────────────┐      │
│ │ Aspect           │ Score    │ Trend (vs Q2)    │      │
│ ├──────────────────┼──────────┼──────────────────┤      │
│ │ Financial Perf   │ +0.42 🟢 │ ↑ +0.12          │      │
│ │ Legal/Regulatory │ -0.85 🔴 │ ↓ -0.20 (⚠️)     │      │
│ │ Operations       │ -0.35 🟠 │ ↓ -0.15          │      │
│ │ ESG/Reputation   │ -0.18 🟠 │ ↔ Stable         │      │
│ └──────────────────┴──────────┴──────────────────┘      │
│                                                          │
│ 🔍 TOP RISK SIGNALS:                                     │
│ • Legal: "Autopilot lawsuit settlement costs rising"     │
│ • Operations: "Battery supply chain delays continue"     │
│                                                          │
│ 📰 NEGATIVE SENTENCES (Legal Aspect):                    │
│ 1. "The Company faces ongoing litigation related to      │
│     Autopilot safety claims..." [Score: -0.92]           │
│ 2. "Material weaknesses in internal controls over..."    │
│     [Score: -0.78]                                       │
│                                                          │
│ 🌐 RELATED ENTITIES (from Knowledge Graph):              │
│ ├─ Panasonic (Supplier): Sentiment -0.45 🟠              │
│ ├─ Tesla Energy (Subsidiary): Sentiment +0.32 🟢         │
│ └─ Rivian (Competitor): Sentiment +0.58 🟢               │
│                                                          │
└──────────────────────────────────────────────────────────┘
```

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• "Supplier Panasonic also negative → supply chain risk confirmed."

BUTTON 2: Sentiment Trends & LSTM Forecasting ✅ TIME-SERIES

What stakeholder sees:

text

┌──────────────────────────────────────────────────────────┐
│ TSLA - Sentiment Over Time (Past 8 Quarters)             │
├──────────────────────────────────────────────────────────┤
│                                                          │
│ [Interactive Plotly Line Chart]                          │
│   • Overall sentiment (black line)                       │
│   • Legal aspect (red line)                              │
│   • Financial aspect (green line)                        │
│   • LSTM forecast next 2 quarters (dotted line)          │
│                                                          │
│ 📈 LSTM Forecast:                                        │
│ Q4 2024 (predicted): -0.38 (70% confidence)              │
│ Q1 2025 (predicted): -0.45 (60% confidence)              │
│                                                          │
│ ⚠️ WARNING: Model predicts continued deterioration       │
│                                                          │
```

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Answer   Links

┌──────────────────────────────────────────────────────────┐
│ Portfolio Optimization Tool                              │
├──────────────────────────────────────────────────────────┤
│                                                          │
│ Strategy: Long-Short Based on Sentiment Scores           │
│                                                          │
│ Parameters:                                              │
│ ├─ Minimum Sentiment (Long): +0.2  [slider]              │
│ ├─ Maximum Sentiment (Short): -0.3 [slider]              │
│ └─ Portfolio Size: Top 20 Long, Top 20 Short             │
│                                                          │
│ [▶ RUN OPTIMIZATION]                                     │
│                                                          │
│ 📊 RESULTS:                                              │
│                                                          │
│ TOP 10 LONG POSITIONS (Highest Positive Sentiment):      │
│ ┌──┬──────┬─────────┬────────┬──────────────┐           │
│ │ # │ Ticker │ Sentiment │ Weight │ Key Strength │      │
│ ├──┼──────┼─────────┼────────┼──────────────┤           │
│ │ 1 │ NVDA   │ +0.78   │ 5.2%   │ Financial ✅ │      │
│ │ 2 │ MSFT   │ +0.65   │ 4.8%   │ All positive │      │
│ │ 3 │ AAPL   │ +0.58   │ 4.5%   │ Operations ✅ │      │
│ │ ...│       │         │        │              │      │
│ │ 10 │ GOOGL  │ +0.42   │ 3.8%   │ Financial ✅ │      │
│ └──┴──────┴─────────┴────────┴──────────────┘           │
│                                                          │
│ TOP 10 SHORT POSITIONS (Most Negative Sentiment):        │
│ ┌──┬──────┬─────────┬────────┬──────────────┐           │
│ │ 1 │ TSLA   │ -0.68   │ -5.0%  │ Legal risk 🔴 │      │
│ │ 2 │ META   │ -0.55   │ -4.7%  │ ESG issues 🔴 │      │
│ │ ...│       │         │        │              │      │
│ └──┴──────┴─────────┴────────┴──────────────┘           │
│                                                          │
│ 📉 EXPECTED PERFORMANCE:                                 │
│ • Sharpe Ratio: 0.88 (vs 0.67 baseline)                  │
│ • Annualized Return: 16.2% (vs 11.8% baseline)           │
│ • Max Drawdown: -18% (vs -24% baseline)                  │
│                                                          │
│ [💾 EXPORT TO CSV] [📊 VIEW BACKTEST]                    │
```
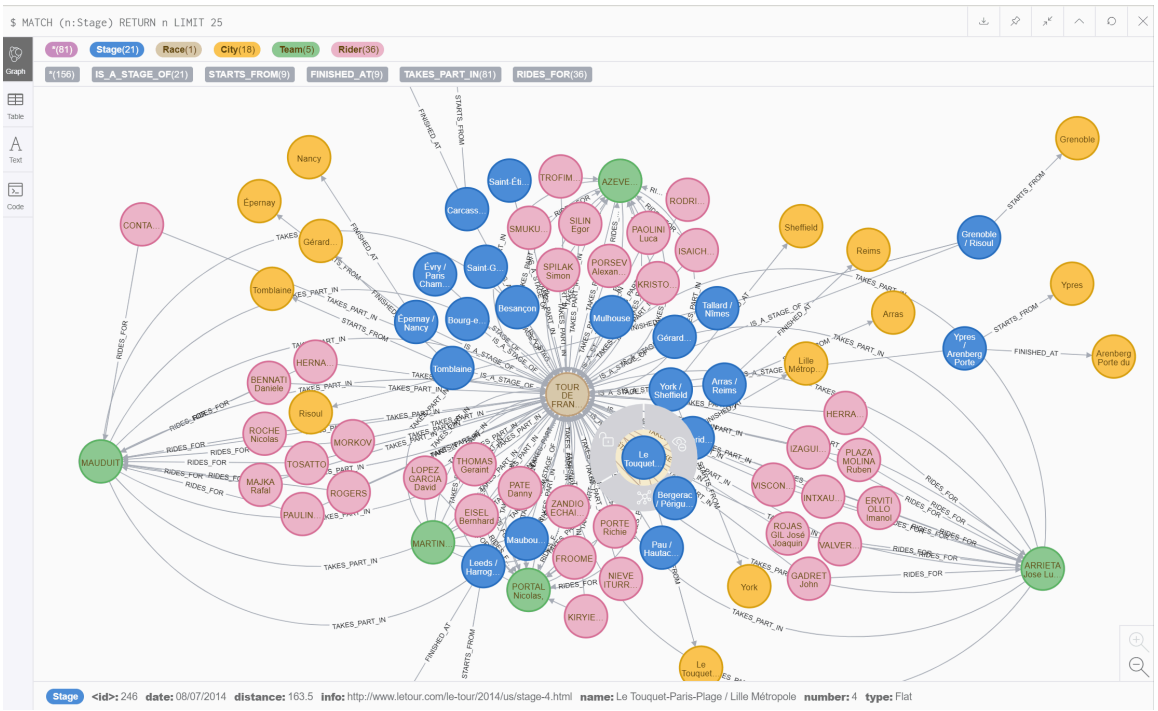
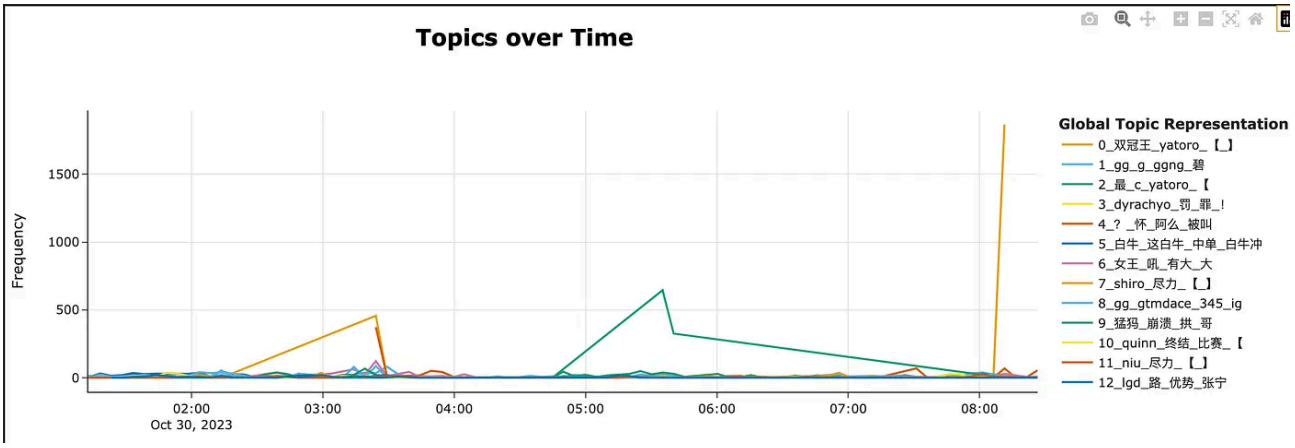
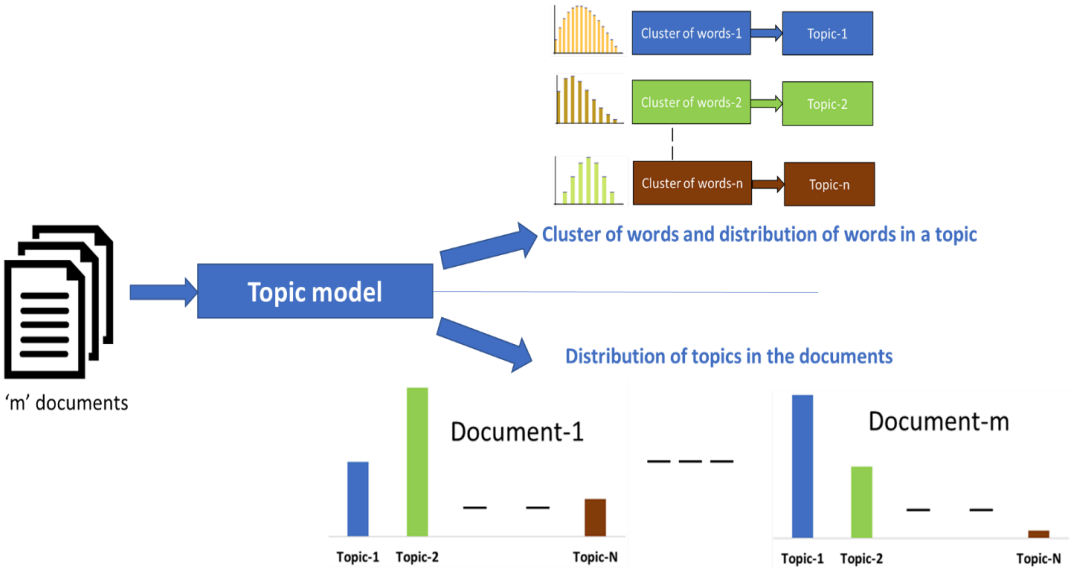

Finance Knowledge Graph (Fig 3)



Topic Modeling Possible Outputs (Fig 4)



https://medium.com/@sleepywyn/a-guide-to-topic-modeling-for-time-series-data-57e59aa695cd



https://github.com/LaurentVeyssier/Topic-Modeling-and-Document-Categorization-using-Latent-Dirichlet-Allocation

Document would be assigned "tags" within application to allow for filtering in search